1 Display your network configuration (ifconfig or ipconfig /all)

```
┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.30.128  netmask 255.255.255.0  broadcast 192.168.30.255
        inet6 fe80::c3ed:d145:e23c:9340  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:b7:30:9b  txqueuelen 1000  (Ethernet)
        RX packets 1501  bytes 275010 (268.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 599  bytes 69328 (67.7 KiB)
        TX errors 0  dropped 4 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2 Show your routing table

```
┌──(kali㉿kali)-[~]
└─$ ip route
default via 192.168.30.2 dev eth0 proto dhcp src 192.168.30.128 metric 100
192.168.30.0/24 dev eth0 proto kernel scope link src 192.168.30.128 metric 100
```

3 Display active network connections

```
┌──(kali㉿kali)-[~]
└─$ ss -tuln
Netid           State           Recv-Q          Send-Q                  Local Address:Port                  Peer Address:Port
```

4 Test connectivity to google.com (ping)

```
┌──(kali㉿kali)-[~]
└─$ ping google.com
PING google.com (142.251.43.14) 56(84) bytes of data.
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=1 ttl=128 time=61.4 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=2 ttl=128 time=59.8 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=3 ttl=128 time=72.5 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=4 ttl=128 time=70.7 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=5 ttl=128 time=73.0 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 59.777/67.460/73.013/5.704 ms
```

5 Trace route to google.com

```
┌──(kali㉿kali)-[~]
└─$ traceroute google.com
traceroute to google.com (142.251.43.14), 30 hops max, 60 byte packets
 1  192.168.30.2 (192.168.30.2)  3.291 ms  2.926 ms  2.735 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

My IP Address

Found in ifconfig

My system IP address is **192.168.30.128**, which is a private IP used within my local network.

Default Gateway

Found in ip route

My default gateway is **192.168.1.1**, which is the router that connects my system to the internet.

DNS Servers

Check using: cat /etc/resolv.conf

My DNS server is 8.8.8.8, which is used to translate domain names like google.com into IP addresses.

Number of Hops to google.com

From traceroute google.com

Count the lines (hops)

It takes approximately **30 hops** to reach google.com from my system.